

Go-Drives System Flow

login/signup.

signup fields :

- guc email
- guc id
- first name
- last name
- major
- password
- photo of id

after approval, email is sent and user can login

car dashboard has multiple screens

- Live Camera Feed
- Lidar data
- Radar data
- Map Screen
- Passenger Data
- Car Data

App Features

- multiple screens
 - Live Camera Feed
 - Lidar data
 - Radar data
 - Map Screen
 - Passenger Data
 - Car Data
- Live Camera Feed
 - front, back, others
 - bounding boxes on nearby objects (pedestrians, vehicles, scene objects)
 - highlight/draw lanes
- Notifications
 - low battery
 - pedestrian crossing

- vehicle passing
- car stopped
- car moving
- passenger drop-off
- passenger coming-in
- accept/deny remote access
- when road is altered
- Passenger data
 - ending the ride
 - passenger authentication (qrcode/pin)
 - manual override if passenger leaves early
 - who will be dropped off in the next stop
- car data
 - battery level
 - speed
 - x- minutes till next stop
- pedestrian, vehicles
 - will show a bounding box around them on screen
- scene
 - highlight lanes and other scene objects if necessary

Milestones

March - Focus on Live Video Streaming

Week 1 (March 18 - March 24)

- Research **live streaming options** (WebRTC, RTSP, or another method)
- Choose the best approach for **low-latency video streaming**
- Set up a **basic test stream** in the mobile app

Week 2 (March 25 - March 31)

- Implement a **working live camera feed** in the app
- Optimize for **latency, stability, and performance**
- Test streaming with **different network conditions**
- Ensure video can be **accessed from multiple devices if needed**

April - ROS Integration & Bounding Boxes

Week 3 (April 1 - April 7)

- Implement **ROS message subscription** in the mobile app
- Ensure the app **receives passenger & pedestrian notifications**
- Research how to **overlay bounding boxes** on live video

Week 4 (April 8 - April 14)

- Implement **bounding box drawing** on the live camera feed
- Test **real-time performance** of bounding box rendering
- Begin implementing **map view with vehicle location**
- Start writing **thesis methodology section**

Week 5 (April 15 - April 21)

- Integrate **finalized route visualization** on the map
- Ensure real-time **sync between ROS messages and the map**
- Conduct **initial testing** of all implemented features
- Write **background and literature review sections** of the thesis

May - Final Testing & Thesis Completion

Week 6 (April 22 - April 28)

- Conduct **full integration testing** in a real or simulated environment
- Make final **UI/UX refinements**
- Implement **error handling & fail-safes**

Week 7 (April 29 - May 5)

- Finalize the **thesis draft** (proofreading, citations, formatting)
- Perform **final bug fixes and optimizations**

Week 8 (May 6 - May 12)

- Submit the **final thesis**
- Prepare for the **presentation & defense**

App Design Trail One

Go Drives

78%



65
mph



Centr
al
Statio
n

Next Stop



12
min



Lidar Data



Car Data



Radar Data



Passenger Data



Camera Views



Map Screen



Admin Dashboard

User Management



Filter

Total Users

Active

Pending

Active

Pending

john@example.com

New York



ACTIVE



jane@example.com

Los Angeles



PENDING



mike@example.com

Chicago



ACTIVE

Sarah Williams





Go Drives

Admin Dashboard



User Management



Car Management

 Filter ▾

38

Total Cars

32

Active

4

Pending

All

Active

Pending

Maintenance



Tesla Model 3

Plate: ABC123

Driver: John Doe



ACTIVE



Toyota Prius

Plate: XYZ789

Driver: Jane Smith



MAINTENANCE



Ford Mustang

Plate: DEF456

Driver: Mike Johnson



PENDING



Chevrolet Bolt





Go Drives

Camera Views

Front Camera

Live



Front view active

Back Camera

Offline



Connection lost

Side 1

Offline